

Employee Attrition Prediction Using Machine Learning

1. Introduction

Employee attrition is an important problem for organizations because employees leaving a company can affect productivity, recruitment costs, training costs, and overall business performance. Predicting employee attrition can help organizations identify employees who are more likely to leave and take appropriate retention measures.

This project develops a machine learning-based Employee Attrition Prediction system. The model uses employee-related information such as age, job role, income, overtime, and other employment-related factors to predict whether an employee is likely to leave the company or stay.

The project follows the major stages of a data science workflow, including data collection, data cleaning, exploratory data analysis, preprocessing, feature selection, model development, evaluation, cross-validation, hyperparameter tuning, and model selection.

2. Problem Statement

The main problem is to predict whether an employee will leave the company or not based on employee-related features.

The project aims to build a machine learning model that can identify employees who are likely to leave the organization.

3. Objectives

The main objectives of this project are:

- To collect and explore employee attrition data.
 - To clean and preprocess the dataset.
 - To analyze important patterns and relationships using EDA.
 - To prepare numerical and categorical features for machine learning.
 - To develop multiple classification models.
 - To evaluate models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
 - To perform cross-validation and hyperparameter tuning.
 - To select the most suitable final model.
 - To save the trained model for future deployment.
 - To develop a user-friendly application using Streamlit as the next deployment stage.
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4. Dataset Description

The project uses an Employee Attrition dataset containing employee demographic, job-related, and organizational information.

The target variable is:

Attrition

- Yes – Employee left the company.
- No – Employee stayed in the company.

For machine learning, the target variable was converted into numerical form:

- Yes = 1
- No = 0

The dataset contains both numerical and categorical variables.

Examples of features include:

- Age
 - Monthly Income
 - Job Role
 - OverTime
 - Job-related information
 - Employee-related demographic information
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5. Data Collection and Exploration

The dataset was loaded using Pandas.

Initial data exploration was performed using:

- shape
- head()
- info()
- describe()
- Missing-value checking
- Duplicate-value checking

These steps were used to understand the structure, data types, statistical characteristics, and possible data-quality problems in the dataset.

The target variable was also analyzed to understand the distribution between employees who stayed and employees who left.

6. Data Cleaning and Transformation

Several preprocessing steps were performed before model training.

6.1 Duplicate Removal

Duplicate records were checked and removed using:

```
df = df.drop_duplicates()
```

6.2 Target Encoding

The categorical target variable `Attrition` was converted into binary numerical values:

```
df["Attrition"] = df["Attrition"].map({  
    "Yes": 1,  
    "No": 0  
})
```

6.3 Feature and Target Separation

The independent variables were separated from the target variable:

```
X = df.drop("Attrition", axis=1)  
y = df["Attrition"]
```

6.4 Categorical and Numerical Features

The features were separated into:

- Numerical features
- Categorical features

Numerical features were standardized using `StandardScaler`.

Categorical features were converted into numerical form using `OneHotEncoder`.

A `ColumnTransformer` was used to apply the appropriate preprocessing to each type of feature.

7. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics and patterns of employee attrition.

The following visualizations were created:

- Attrition count plot
- Age distribution
- Monthly Income distribution
- Overtime vs Attrition
- Job Role vs Attrition

Key EDA Findings

The analysis showed that employee attrition is not equally distributed between the two target classes. The number of employees who stayed is higher than the number of employees who left.

The analysis also explored the relationship between overtime, job role, age, and income with employee attrition.

These visualizations helped understand which employee-related characteristics may be associated with attrition.

8. Class Imbalance

The target variable is imbalanced because the number of employees who stayed is considerably higher than the number of employees who left.

For example, in the final test set:

- Class 0 (No Attrition): 247
- Class 1 (Attrition): 47

Because attrition is the minority class, accuracy alone is not sufficient to judge the model.

Therefore, Precision, Recall, F1-score, and ROC-AUC were also considered.

During hyperparameter tuning, class weighting was considered for Logistic Regression and Random Forest using:

```
"class_weight": [None, "balanced"]
```

SMOTE was not used in the final implementation.

9. Feature Selection

The target variable `Attrition` was separated from the remaining features.

The available employee-related features were used for model development after preprocessing.

Categorical features were encoded using One-Hot Encoding, while numerical features were standardized.

The preprocessing was integrated into machine learning pipelines to ensure consistent transformation during training and testing.

10. Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio.

The following approach was used:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

Stratification was used to maintain a similar class distribution in the training and testing datasets.

11. Model Development

Several machine learning classification algorithms were trained and evaluated.

The models included:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. AdaBoost
5. Gradient Boosting
6. XGBoost

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

12. Initial Model Evaluation

The initial test results were:

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.864	0.706	0.255	0.375	0.618
Decision Tree	0.779	0.327	0.362	0.343	0.610
Random Forest	0.861	0.667	0.255	0.369	0.616
AdaBoost	0.830	0.412	0.149	0.219	0.554
Gradient Boosting	0.854	0.600	0.255	0.358	0.612
XGBoost	0.844	0.519	0.298	0.378	0.623

The initial models showed relatively low recall for the attrition class. Therefore, further cross-validation and hyperparameter tuning were performed.

13. Cross-Validation and Hyperparameter Tuning

Five-fold Stratified Cross-Validation was used during hyperparameter tuning.

The following models were selected for further tuning:

- Logistic Regression
- Random Forest
- XGBoost

`GridSearchCV` was used to find suitable hyperparameter combinations.

The tuning process used F1-score as the scoring metric because the dataset contains class imbalance and identifying the minority attrition class is important.

14. Tuned Model Results

After hyperparameter tuning, the following results were obtained:

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.7551	0.3529	0.6383	0.4545	0.7658
Random Forest	0.8503	0.5556	0.3191	0.4054	0.7845
XGBoost	0.8639	0.7059	0.2553	0.3750	0.7691

15. Final Model Selection

Logistic Regression was selected as the final model.

Although XGBoost achieved the highest accuracy and Random Forest achieved the highest ROC-AUC, Logistic Regression achieved the highest F1-score and recall among the tuned models.

The final Logistic Regression model achieved:

- **Accuracy:** 75.51%
- **Precision:** 35.29%
- **Recall:** 63.83%
- **F1-score:** 45.45%
- **ROC-AUC:** 76.58%

Recall was particularly important because the objective is to identify employees who are likely to leave the company.

Therefore, Logistic Regression provides a better balance for identifying potential attrition cases.

16. Final Model Evaluation

The final Logistic Regression model was evaluated on the unseen test dataset.

The classification report was:

Class	Precision	Recall	F1-score	Support
0 – No Attrition	0.92	0.78	0.84	247
1 – Attrition	0.35	0.64	0.45	47

Overall accuracy was **75.51%**.

Confusion Matrix

The confusion matrix was:

```
[[192, 55],  
 [ 17, 30]]
```

This means:

- True Negative = 192
- False Positive = 55
- False Negative = 17
- True Positive = 30

The model correctly identified **30 out of 47 employees who belonged to the attrition class**.

This gives an attrition recall of approximately **64%**, which is useful because the model is able to identify a significant portion of employees who may leave.

17. Model Saving

After selecting the final Logistic Regression model, the complete pipeline was saved using Joblib.

```
import joblib  
  
best_model = lr_grid.best_estimator_  
  
joblib.dump(  
    best_model,  
    "employee_attrition_model.pkl"  
)
```

The saved model contains the preprocessing and trained model pipeline, which makes it suitable for later use in the application.

18. Model Deployment

The project guideline requires the model to be deployed through a user-friendly application such as Streamlit.

The trained model can be loaded using:

```
import joblib

model = joblib.load("employee_attrition_model.pkl")
```

The planned Streamlit application will allow users to enter employee information and receive an attrition prediction.

The application will provide a simple interface where employee-related attributes can be entered and the trained model can predict whether the employee is likely to leave the company.

19. GitHub and Version Control

GitHub is used to maintain and organize the project files.

The project repository is:

 [url](https://github.com/Ashishjoshi794/Employee-Attrition-Prediction) Employee Attrition Prediction – GitHub Repository  <https://github.com/Ashishjoshi794/Employee-Attrition-Prediction> 

The repository is intended to contain the project notebook, documentation, model-related files, and deployment files.

20. Limitations

The project has several limitations:

- The dataset contains an imbalanced target variable.
 - The model does not guarantee that an employee will actually leave.
 - The model performance depends on the available features and dataset.
 - The dataset may not represent every organization or industry.
 - Some false positive and false negative predictions are still present.
 - SMOTE was not used in the final implementation.
 - Further testing with additional datasets could improve confidence in generalization.
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21. Future Improvements

The project can be improved in the future by:

- Testing SMOTE and comparing it with class weighting.
 - Trying additional machine learning algorithms.
 - Performing more advanced feature engineering.
 - Optimizing the prediction threshold to improve attrition recall.
 - Adding more employee-related features.
 - Developing a complete Streamlit application.
 - Deploying the Streamlit application online.
 - Adding interactive visualizations and dashboards.
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22. Conclusion

This project developed a machine learning system for predicting employee attrition.

The dataset was cleaned and explored, followed by preprocessing of numerical and categorical features. Several classification algorithms were trained and compared. Cross-validation and GridSearchCV were used for hyperparameter tuning.

Among the tuned models, Logistic Regression was selected as the final model because it achieved the highest F1-score and recall. The final model achieved an accuracy of **75.51%**, recall of **63.83%**, F1-score of **45.45%**, and ROC-AUC of **76.58%**.

The model can help organizations identify employees who may be at higher risk of leaving and can support HR teams in making better employee retention decisions.

The trained model has also been saved using Joblib and can be integrated into a Streamlit application for interactive prediction.

23. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost
- Joblib
- Jupyter Notebook

- Git
 - GitHub
 - Streamlit
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24. Project Workflow

```
Data Collection
  ↓
Data Exploration
  ↓
Data Cleaning
  ↓
EDA
  ↓
Feature Preparation
  ↓
Train-Test Split
  ↓
Data Preprocessing
  ↓
Model Development
  ↓
Model Evaluation
  ↓
5-Fold Cross-Validation
  ↓
Hyperparameter Tuning
  ↓
Model Comparison
  ↓
Final Logistic Regression Model
  ↓
Model Saving
  ↓
Streamlit Deployment
```

25. Final Summary

The project successfully demonstrates the complete machine learning workflow for employee attrition prediction, from data exploration and preprocessing to model development, evaluation, hyperparameter tuning, and final model selection.

The selected Logistic Regression model provides a reasonable solution for identifying potential employee attrition cases, particularly because recall and F1-score were prioritized during model selection.